PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

AgriAid+: AI-Powered Smart Farming & Climate Resilience Platform

Group No.1

Sharayu Mahajan (22107051)
Rishi Mane (22107063)
Kalpana Mohanty (22107059)
Monish Mudaliar (22107027)

Project Guide
Dr. Veena Trivedi
Co-Guide: Prof. Aishwarya Londhe

Contents

- Abstract
- Introduction
- Objectives
- Literature Review
- Research Gap
- Problem Definition
- Scope
- Technological Stack
- Proposed System Architecture/Working/Module Diagram
- Prototype Design Demonstration
- Implementation Status
- Review Suggestions (Given in Last meeting)

Abstract

Agriculture in India faces critical challenges such as unpredictable climate conditions, limited access to scientific crop and fertilizer recommendations, language barriers in advisory services, and market inefficiencies that reduce farmer profitability. To address these, our project AgriAid+ proposes an AI-powered smart farming and climate resilience platform that integrates four key modules. The system includes a Crop and Fertilizer Recommendation Engine that leverages soil, weather, and regional datasets for personalized guidance, a Multilingual Farmer Advisory Chatbot enabling easy communication in local languages, Weather Forecasting and Market Price Trend Analysis to support better planning and sales strategies, and a Farmer-Centric Marketplace Module that promotes direct trading of produce and equipment. By combining AI, natural language processing, and real-time analytics, AgriAid+ empowers farmers with actionable insights, enhances productivity, ensures fair pricing, and supports sustainable and climate-resilient agriculture.

Introduction

Agriculture is the backbone of India's economy and the primary source of livelihood for millions in rural areas. Yet, farmers continue to struggle with everyday challenges. Many still lack proper, data-driven guidance on which crops or fertilizers to use. Unpredictable weather and climate shifts often result in heavy crop losses. On top of that, market inefficiencies and dependence on middlemen prevent farmers from earning fair prices for their produce. Accessing modern farming practices or government schemes is also difficult, especially due to language barriers. While tools like weather apps, eNAM, and government portals exist, they are often fragmented, not personalized, and rarely available in regional languages. This makes it clear that farmers need a single, AI-powered platform that provides them with real-time, localized, and practical solutions to support better decisions and improve their livelihoods.

Objectives

- To develop a personalized crop and fertilizer recommendation system using soil data, leveraging advanced machine learning algorithms such as XGBoost and LightGBM.

- To design a multilingual farmer advisory chatbot that enables seamless communication and support in local languages.

- To integrate weather forecasting and market price analytics for better crop planning and income security.

- To build a farmer-centric digital marketplace for direct trading of produce and agricultural equipment.

Sustainable Development Goals(SDG) mapped

SDG no. : 13

SDG description: Climate Action

How it is mapped:
Weather forecasting and climate-aware crop planning help farmers adapt to changing climatic conditions, reducing crop loss and supporting climate resilience in agriculture.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
1	Advancing crop recommendation system with supervised machine learning and explainable artificial intelligence	Sourabh Shastri, Sachin Kumar, Vibhakar Mansotra & Rohit Salgotra	2025	Supervised machine learning (Gradient Boosting, KNN, DT, RF, NB, SVM, LR, NN, LDA, QDA) and Explainable AI (LIME)	The model's excellent performance on the limited, single Kaggle dataset may not generalize well to varied real-world agricultural conditions.
2	Intent Aware Context Retrieval for Multi-Turn Agricultural Question Answering	Abhay Vijayvargia, Ajay Nagpal, Kundeshwar Pundalik, Smita Gautam, Atharva Savarkar, Viraj Thakur, Pankaj Singh, Rohit Saluja, Ganesh Ramakrishnan	2025	IFT model refined with fine-tuning on Indian agricultural knowledge, multi-turn conversation flow, Retrieval-Augmented Generation (RAG)	The model is currently specialized in only two crops: onions and grapes. Expanding support to a wider variety of crops requires significant dataset expansion and re-engineering of the intent-slot mapping and RAG module.
3	Implementation, Impact, and Challenges of eNAM in Agricultural Markets – A Systematic Literature Review (SLR)	Sushil Chauhan & Shivali Rathore	2025	Systematic Literature Review (SLR)	The review's scope is limited to English, peer-reviewed journal articles from 2016-2024, potentially omitting other relevant research.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
4	Machine Learning in Sustainable Agriculture: Systematic Review and Research Perspectives	Juan Botero-Valencia, Vanessa García-Pineda, Alejandro Valencia-Arias, Jackeline Valencia, Erick Reyes-Vera, Mateo Mejia-Herrera, Ruber Hernández-García	2025	PRISMA-based Systematic Review; Bibliometric Analysis of ML in agriculture (2007–2025, 124 studies)	Reliance on selected databases and limited bibliometric tools may introduce biases and restrict comprehensiveness.
5	Digital Agro-Advisory Tools	Kingsley Ofosu-Ampong, Wuletawu Abera, Tewodros Mesfin	2025	Review of 19 platforms; adoption & behavioural analysis.	Relies on online use; low adoption in rural areas.
6	Hybrid SV-PSO-BiLSTM	Satish Kumar, Amit Verma, Pankaj Kumar, Rakesh Majhi	2025	Hybrid model for agri-price prediction with PSO & BiLSTM.	High complexity & time; limited to prices.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
7	Agro-Digital Tool Adoption	Kingsley Ofosu-Ampong, Tsion Abate, Wuletawu Abera	2025	Categorised tools by adoption (low/med/high).	Limited data; findings not generalisable.
8	ML for Climate-Smart Agriculture	R. Kumar, S. Singh, P. Sharma	2024	Conceptual ML approach for CSA adoption & productivity.	Lacks long-term local climate guidance.
9	Data-driven irrigation optimization	Imade	2024	RF with soil & weather data cut water use by ~25%	Costly sensors, low generalizability.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
10	ML for yield prediction in smart farming	Abourabia Imade, Ounacer Soumaya, Elghoumari Mohammed yassine, Azzouazi Mohamed	2024	Compared regressors; AdaBoost lowest error, RF R^2≈0.89; DL image models ≈99%.	Region specific , data quality issues
11	Smart agriculture: ML & DL for drought stress	TariqAli1, Saif Ur Rehman, Shamshair Ali, Khalid Mahmood, Silvia Aparicio Obregon, Rubén Calderón Iglesias, Tahir Khurshaid & Imran Ashraf	2024	Used SVM, GB, RNN, LSTM on plant protein features; accuracy up to 97%.	Limited field validation, high computation.
12	Predicting climate smart agriculture (CSA) practices using machine learning: A prime exploratory survey	Freddy Noma , Suresh Babu	2024	Optimized Gradient Boosting ML algorithm trained and tested using household-level data from Rakai district in Uganda, framed within climate adaptation analytical frameworks.	The model achieved only up to 60% accuracy, indicating a significant portion (40%) of misclassifications, which the authors suggest could be improved with deep-learning algorithms in future research.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
13	Climate Services for Groundnut Farmers	R. Lopes, L. Blondiaux, D. Diasso, A. Yuga	2024	Micro/meso/macro classification of farmers; ~57–60% accuracy.	Micro-level data too noisy; low accuracy.
14	Smart Farming: Crop Recommendation using Machine Learning with Challenges and Future Ideas	Devendra Dahiphale, Pratik Shinde, Koninika Patil, Vijay Dahiphale	2023	Machine learning algorithms trained and tested on a Kaggle dataset containing soil composition and climate data.	The model's excellent accuracy on a specific Kaggle dataset may not translate to diverse real-world agricultural conditions.
15	Soil Classification & Crop Rec.	Hitiksha Karre, C. Prakash, P. Balamurugan	2020	Bayesian models for soil class & crop choice.	Focus only on soil & crops; No weather link

Research Gap(Limitations of existing systems)

- Most chatbots speak in only one or two languages, leaving out millions of farmers who rely on local dialects to access knowledge.
- Platforms like eNAM still struggle with trust — without ratings or feedback, farmers hesitate to depend on them for critical market decisions.
- Small and noisy datasets weaken the reliability of predictions, often making recommendations less useful in real-world fields.
- Current systems often ignore weather and climate factors, even though these are the biggest forces shaping agriculture today.
- Advanced ML models exist, but they demand heavy computation, keeping powerful solutions out of reach for everyday users.
- Long-term climate data is missing, leaving systems unprepared to guide farmers in adapting to changing weather patterns.

Problem Definition

Challenges Faced by Farmers – Indian farmers struggle with lack of data-driven crop/fertilizer decisions, unpredictable climate risks, limited access to real-time weather/market data, and language barriers in understanding advisories and schemes.

Limitations of Current Solutions – Existing platforms and government initiatives are fragmented, offer minimal personalization, and lack integration of essential agricultural services.

Proposed System: AgriAid+ – A unified AI-powered platform integrating crop and fertilizer recommendations, multilingual advisory chatbot, weather forecasting, market price analytics, and a farmer-centric marketplace.

Scope

- To develop a crop and fertilizer recommendation system that uses soil data, weather data, and agricultural datasets to provide personalized guidance.
- To build a multilingual farmer advisory chatbot that enables farmers to interact in local languages and access information on crop care, government schemes, and best practices.
- To integrate weather forecasting and market price trend analysis for supporting better crop planning, harvesting, and selling strategies.
- To design a farmer-centric marketplace module that allows direct trading of produce and agricultural equipment, ensuring transparency and fair pricing.
- To provide a unified AI-powered platform that combines decision support, real-time insights, and marketplace services to empower farmers and promote sustainable agriculture.

Technological Stack

- HTML5: Used to structure the web portal and design farmer-friendly forms and chatbot interface (Current version: HTML5, W3C standard).

- CSS3: Provides responsive design, regional language font support, and mobile-first layouts (Current version: CSS3).

- JavaScript (ES2023/ES14): Enables interactivity, chatbot UI, and real-time updates using APIs (Current version: ECMAScript 2023).

- Python 3.12: Backend engine for AI/ML, running models for recommendations, forecasting, and NLP chatbot (Current version: Python 3.12, released Oct 2023).

Proposed system architecture

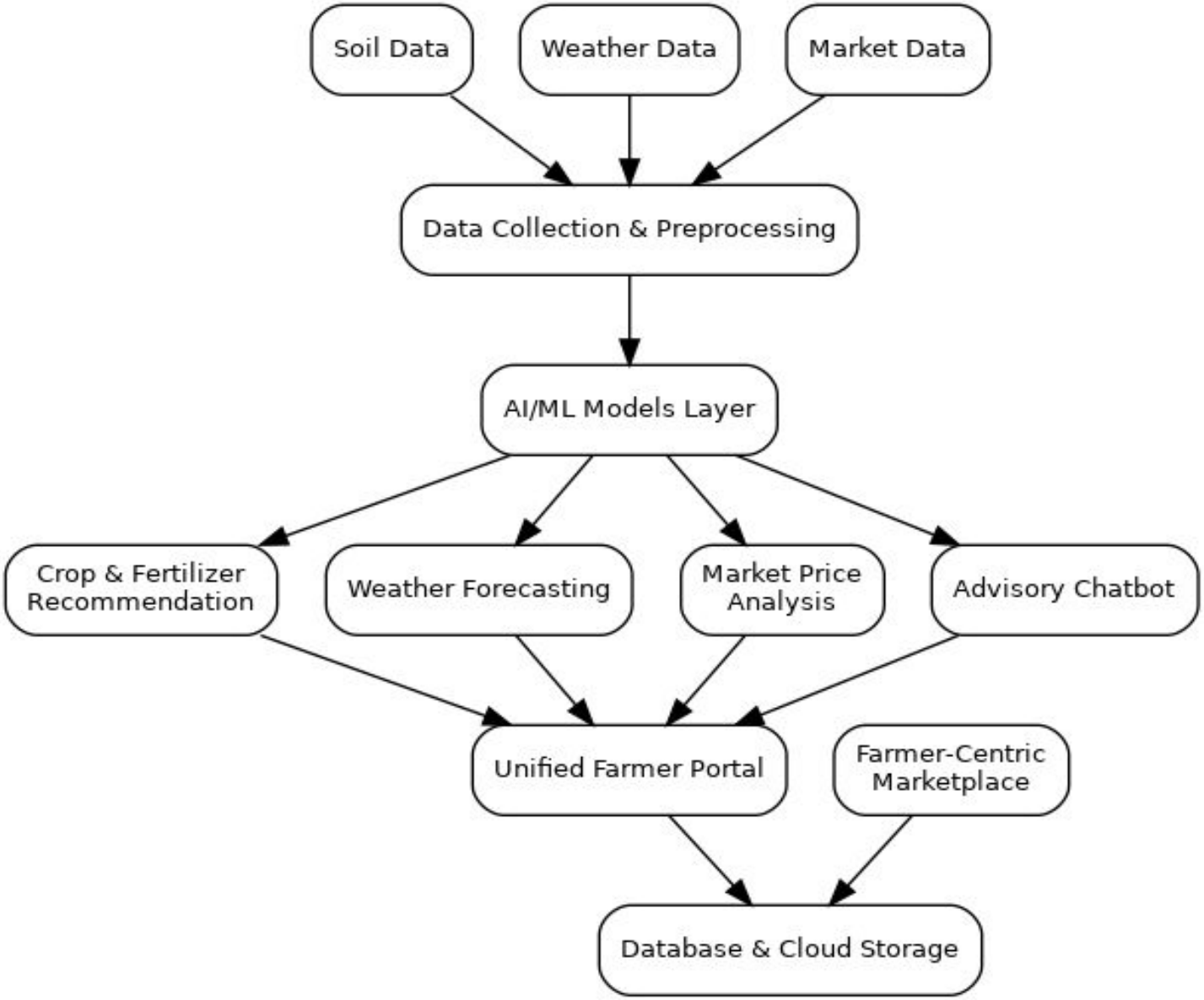

Fig 1: Architecture design for Agridaid+

Implementation Status

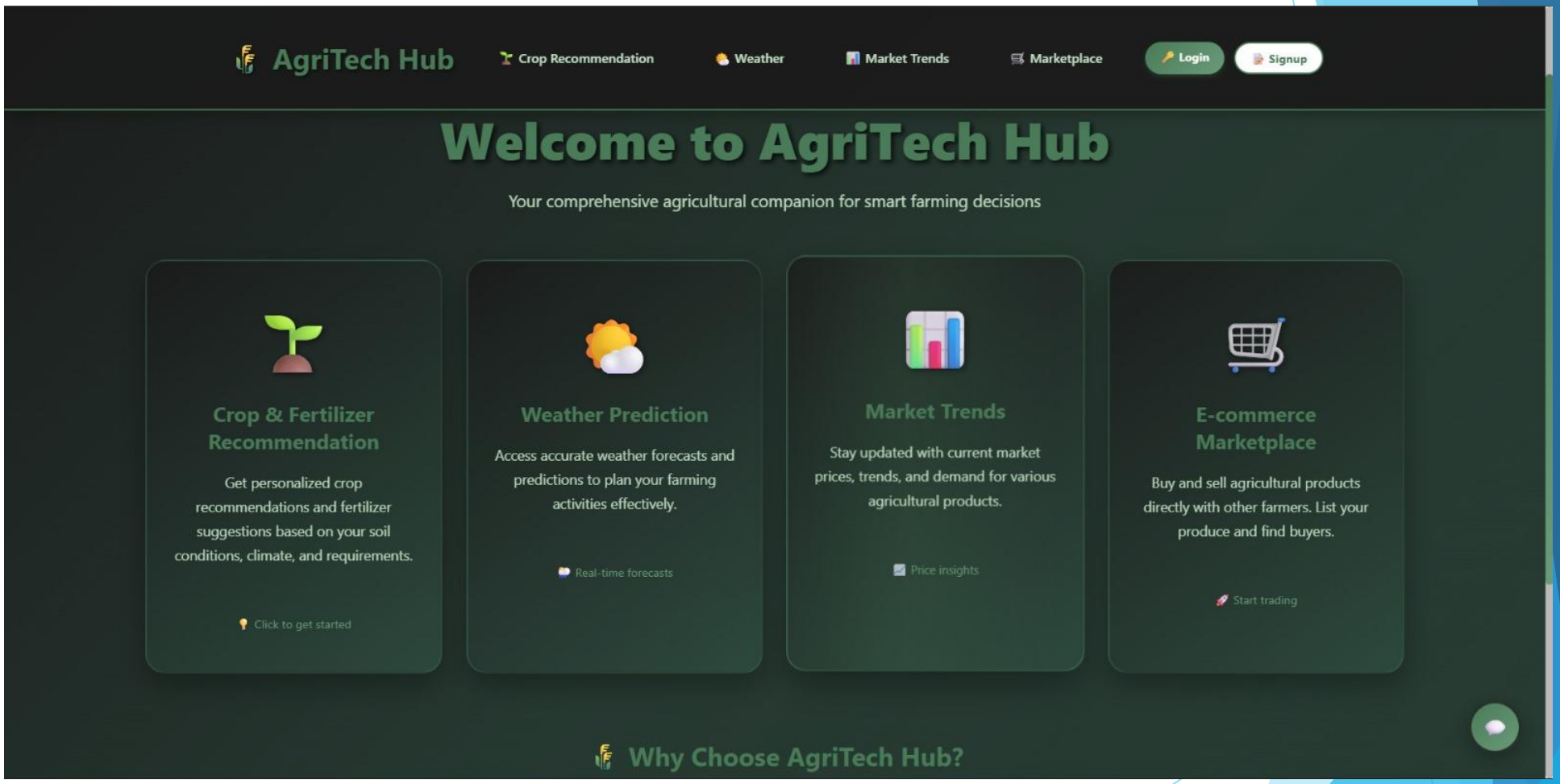

Fig 1: Home Page for Agridaid+

Implementation Status

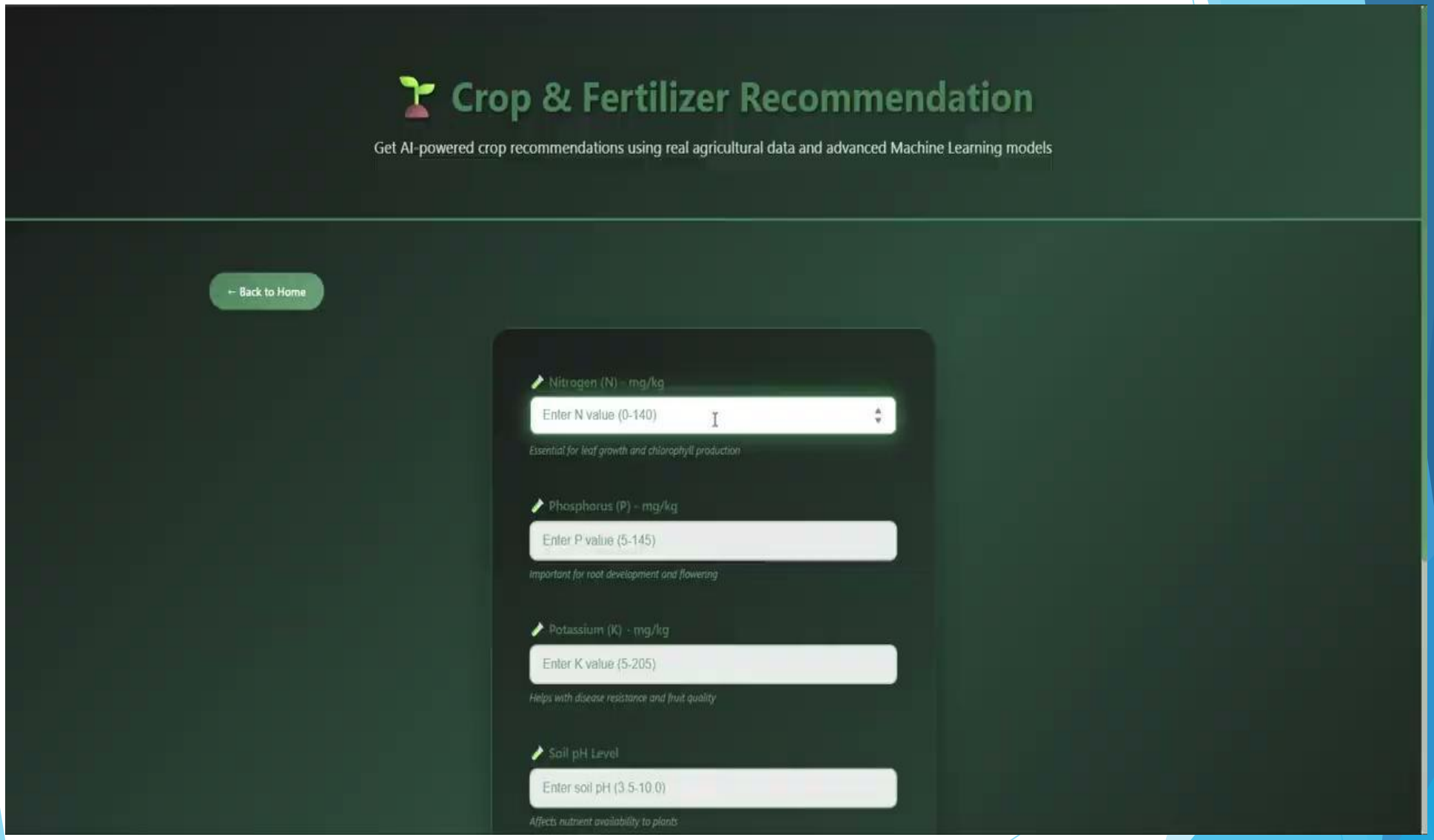

Fig 2: Crop and Fertilizer Recommendation design

Implementation Status

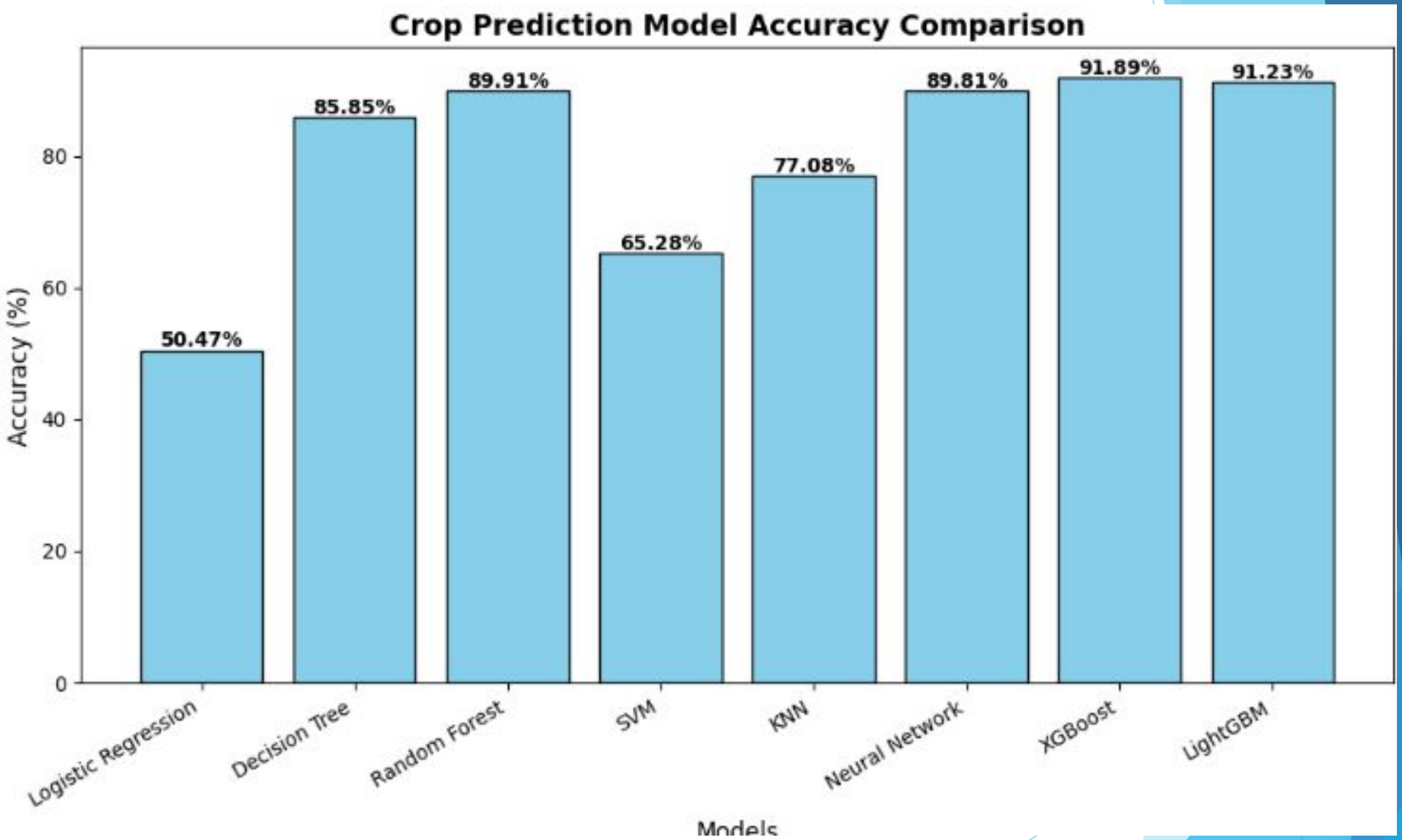

Fig 3 :Comparative Study of ML Algorithms for Crop Classification

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Thank You...!!